

2024 Graduation Thesis Abstract

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Title	Development of a CAM System for Incremental Forming of CFRTP from 3D CAD Data

Purpose of Research:

Due to the increasing demand for lightweight, resilient, and cost-effective materials such as carbon fiber-reinforced thermoplastics in various industries, there is a need to manufacture low-volume, high-complexity products using these materials. Often, these types of products are too costly and inefficient to produce using traditional molding techniques. Incremental forming is an efficient and cost-effective solution that eliminates the need for molds typically found in traditional manufacturing methods, making incremental forming ideal for rapid prototyping and small-batch production using materials such as CFRTP.

The purpose of this research is to enhance a Python-based CAM system that generates precise, error-free toolpaths for the single-point incremental forming of complex shapes using discontinuous CFRTP.

The three primary objectives of the research are to enhance the CAM system's capability to generate effective toolpaths for complex geometries, streamline the design-to-manufacturing process, and demonstrate the effectiveness of the generated toolpaths for manufacturing complex shapes.

Experimental Procedure:

The updated CAM system processes 3D CAD data from STL files generated in Creo Parametric 9.0, extracting point clouds to create an optimized G-code toolpath for testing on a Sodick MC430L CNC machining center. Key modules include point cloud extraction, cluster prevention, overlap prevention, smoothing, and reverse offset to account for tool geometry and material springback. The system utilizes Python libraries, including Pandas, NumPy, and SciPy's ConvexHull, for efficient data handling and toolpath optimization. Experimental conditions involve forming discontinuous CFRTP (20% carbon fiber, 80% nylon 6) and nylon sheets at 120°C and 150°C in a T-22H oil bath with Shin-Etsu KF968 silicone oil, using a SUS304 mandrel (5 mm diameter). Tests were conducted with and without molds (3D-printed onyx and aluminum) to evaluate toolpath performance, with variables such as tool diameter, layer thickness, and minimum point distance adjustable to optimize forming for complex geometries (a 30 mm x 30 mm x 9.9 mm test shape with chamfered and rounded sides).

Results:

The updated CAM system significantly improved over its predecessor, reducing toolpath generation time from 2.5 hours to 2 seconds for a complex test shape, achieving a 99% time reduction through point filtering and convex hull optimization. NCviewer.com simulations confirmed error-free, smooth toolpaths without intersections. Nylon sheet tests validated toolpath reliability, with mold-assisted tests achieving more consistent forming depths and less springback compared to non-molded tests. CFRTP tests at 150°C were more successful than at 120°C, achieving acceptable forming depths but with rough surfaces and minor defects. Tests conducted at 120°C failed due to premature material removal, resulting in deformation and cracking of the material. Limited CFRTP tests (four total) were conducted due to long cooling times, which restricted comprehensive data collection. Additionally, the limitations of the 3D scanner prevented a detailed geometric analysis. The results highlight the CAM system's capability to handle complex geometries and streamline manufacturing, though further research is needed to optimize CFRTP forming conditions and improve analysis methods.

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